

EECS 622: Homework #25

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Problem 1

A receiver is built with a narrowband IF filter at **200 MHz**. The local oscillator is tuned to **800 MHz**. The receiver has no preselector filter.

- A. Say just a single signal appears at the receiver input, at a frequency of **1800 MHz**. Considering up to and including 3rd order products, determine the frequencies of all signals that appear at the IF output of the mixer. Which of these signals will also appear at the detector?

Solution:

There are the mixing products and its spurs which appear at this IF output. *If there were more incoming signals, there would be an image and murphy spectrum which have spurs that could appear too.*

$$f_{IF} = |f_{RF} - f_{LO}| = 1000 \text{ MHz} \quad (1)$$

$$f_{IF} = |f_{RF} + f_{LO}| = 2600 \text{ MHz} \quad (2)$$

$$f_{1a}^{spur} = f_{RF} = 1800 \text{ MHz} \quad (3)$$

$$f_{1b}^{spur} = f_{LO} = 800 \text{ MHz} \quad (4)$$

$$f_{2a}^{spur} = 2f_{RF} = 3600 \text{ MHz} \quad (5)$$

$$f_{2b}^{spur} = 2f_{LO} = 1600 \text{ MHz} \quad (6)$$

$$f_{3a}^{spur} = 3f_{RF} = 5400 \text{ MHz} \quad (7)$$

$$f_{3b}^{spur} = |2f_{RF} - f_{LO}| = 2800 \text{ MHz} \quad (8)$$

$$f_{3d}^{spur} = 2f_{RF} + f_{LO} = 4400 \text{ MHz} \quad (9)$$

$$f_{3e}^{spur} = |f_{RF} - 2f_{LO}| = \boxed{200} \text{ MHz} \quad (10)$$

$$f_{3f}^{spur} = f_{RF} + 2f_{LO} = 4400 \text{ MHz} \quad (11)$$

$$f_{3g}^{spur} = 3f_{LO} = 2400 \text{ MHz} \quad (12)$$

The filter kills off any signal not at 200 MHz, so we just get the spur through it. Everything else appears just before the filter.

- B. Now determine the frequency of any and all signals at the receiver input that will result in some signal power reaching the detector. Consider products up to and including 3rd order.

Solution:

We know that there will definitely be an image signal:

$$\begin{aligned} |f_{RF}^s - f_{RF}^{image}| &= 2f_0^{IF} \\ f_{RF}^{image} &= f_{RF} \pm 2f_0^{IF} \\ f_{RF}^{image} &= \{1600 \text{ MHz}, 2000 \text{ MHz}\} \end{aligned}$$

And some RF signals will produce spurs at the IF frequency. Solving for f_{RF} , starting with the general form of spurrious signals which we have derived,

$$\begin{aligned} f_{spur} &= |mf_{RF} \pm nf_{LO}| \\ mf_{RF} &= \pm f_{spur} \mp nf_{LO} \\ f_{RF} &= \frac{|f_{spur} \mp nf_{LO}|}{m} \\ f_{RF} &= \frac{|200 \text{ MHz} \mp n(800 \text{ MHz})|}{m} \end{aligned}$$

Third order products exist for all $n, m \in \{0, 1, 2, 3\}$. In no particular order, the resulting values are:

$$\{66.67, 100.0, 200.0, 300.0, 333.33, 466.67, 500.0, 600.0, 700.0, 733.33, 866.67, \\ 900.0, 1000.0, 1100.0, 1300.0, 1400.0, 1800.0, 2200.0, 2600.0\} \text{ MHz}$$

And, all of these should be the murphy spectrum signals.

Problem 2

An engineer is designing a single-conversion receiver to work over an RF bandwidth between 1.0 GHz and 1.1 GHz .

This designer uses a single preselector filter, and a single IF filter. This preselector filter provides more than 50 dB of attenuation to all signals with frequencies greater than 1.5 GHz .

This preselector filter likewise provides more than 50 dB of attenuation to all signals with frequencies less than 0.5 GHz .

The receiver designer must achieve at least 50 dB of image rejection.

Determine:

1. the lowest possible Intermediate Frequency (f_0^{IF}) for this design, as well as the required LO frequencies, if high-side tuning is used.

Solution:

We need to bring our image band below 0.5 GHz, so that it is attenuated. Our image bandwidth is related to our choice in IF, such that:

$$|f_{RF}^s - f_{RF}^{image}| = 2f_0^{IF}$$

We may simply set the image to 0.5 GHz for the largest RF signal to determine the IF,

$$f_0^{IF} = \frac{1}{2}|1.1 - 0.5| = 0.3 \text{ GHz}$$

Where, the required LO frequencies (for high-side tuning) will be:

$$\begin{aligned} 1.0 + 0.3 \text{ GHz} &\leq f_{LO} \leq 1.1 + 0.3 \text{ GHz} && (\text{since } f_{LO} = f_{RF}^s + f_0^{IF} \text{ for high-side}) \\ 1.3 \text{ GHz} &\leq f_{LO} \leq 1.4 \text{ GHz} \end{aligned}$$

2. the lowest possible Intermediate Frequency (f_0^{IF}) for this design, as well as the required LO frequencies, if low-side tuning is used.

Solution:

IF frequency will remain the same, but LO frequency will become:

$$\begin{aligned} 1.0 - 0.3 \text{ GHz} &\leq f_{LO} \leq 1.1 - 0.3 \text{ GHz} && (\text{since } f_{LO} = f_{RF}^s - f_0^{IF} \text{ for low-side}) \\ 0.7 \text{ GHz} &\leq f_{LO} \leq 0.8 \text{ GHz} \end{aligned}$$